

ActInf Livestream #005.2 ~ "Multiscale integration: beyond internalism and

Livestream | (2019) | 1:34:55

hello

everyone welcome to the active inference

live stream

this is act imp stream 5.2 it is

october 6 2020.

first time listeners or not welcome to

team com

we are an experiment in online team

communication and learning related to

active inference

you can find us on twitter at

inferenceactive

you can email us at activeinference

gmail.com

on keybase at our public team or

on youtube this is a recorded and an

archive live stream so please provide us

feedback live participants as well as

viewers so that we can improve our work

all backgrounds and perspectives are

welcome here and

also just as far as video etiquette goes

please mute

if there's noise in your background and

raise your hand so that we can use the

stack to hear from everyone

as mentioned we are here in act imp

stream 5.2 and here's a little bit about

how today is going to go

first we're going to have our regular

warm-up section where we'll do some

intros and we'll check in with a few

questions

last time in active stream 5.1 we did

a pretty broad overview summary of the

paper went through the figures the goals
the abstract
and today in the follow-up discussion
part two
we're going to be going through first
off any
questions or comments that people want
to address and then after we've
exhausted that or if we're excited just
to get to
the other topics we'll go right ahead we
will
return to the debate about internalism
versus externalism because that is in
some ways what the
broader scholastic tradition that the
paper is involved in
is about we'll talk about markov
blankets
we'll talk about what it means to be
internal and what it means to be
generative
we'll talk about changing one's mind
we'll talk about free energy
minimization we'll talk about some of
the limits of cognition and science
and then we'll hopefully return to the
figures especially figure 6
and mechanistic differences in different
multi-scale systems
so here we are in the introduction
section
for this intro section please introduce
yourself and your location
just say a quick hello and a short
introduction if you would like
and then pass it to someone else i'll go
first

i'm daniel friedman i'm in davis right
now
and i will pass it to stephen
hello i'm stephen i'm in toronto right
now and
uh very excited about being here today
i'm going to pass it to alex
hello i'm alex and today i'm in moscow
as well as usually
hello everybody and they pass it to one
hi my name is ivana i'm from moscow also
and i passed to alejandra
hello hello everybody my name is
alejandra and
also as usual i'm here in mexico
and i pass it to sasha
hi good morning everybody um i am as
usual
in davis california and i'm looking
forward to this conversation
and i pass it to maxwell
hey everyone i am maxwell ramstad happy
to be
joining this conversation from sunny
montreal in canada
and i'll pass it to shannon
hi everyone i'm shannon and i'm
currently in south dakota
awesome does that get through everyone
yep
a nice small tight group today thanks
everyone for
coming in so let's go through our
warm-up questions
and we would love to hear from all of
you on these
the first warm-up question is what got
you excited for today's discussion what

brought you to the follow-up
discussion and people can raise their
hand
all start what got me excited was
that second phase of interacting with
the paper which is beyond
just a first pass what was said in the
paper and what the figures represented
to tie it to deeper questions and to
deeper ideas
yeah stephen yeah and what really
excites me
well there's a lot of stuff that came up
last week so that was really useful and
i got some new
insights but i'd say um i saw a
potential
way that it was moving towards this
multi-method potential to bring together
different ways of thinking
and practices so um i'm
i'm interested in how the kind of
different
levels sort of can come together in
different ways
and privilege different types of um
scales in ways which are actually quite
practical and useful
so that was that was really really
interesting
cool and we'll definitely talk about how
the different
scales come to bear if anyone else wants
to raise their hand otherwise i'll just
go to the second discussion question
which is
if at all how have you updated your
beliefs on multi-scale integration

since the last conversation i think that
was a
question from sasha so maybe i'll
go to her while other people are raising
their hand
while she unmutes um
yeah that was you know a bit of a cheeky
question um
but i think we got into a lot of
interesting topics last week
um uh you know
and really getting into um the
philosophy of science
um that was really interesting for me
and
uh it's a topic i hope to explore
especially in thinking about
um different levels of um
how to create the appropriate system for
studying cognition
and where to draw that boundary
um something i'm interested for
uh figuring out in my own work with
studying neuroscience and development
and um just really enjoyed hearing
everyone's perspective as well
so my uh my update was that um
philosophy of science is really critical
to uh
the practic practicality of experimental
design
yeah there's in many of these papers and
ideas there's us experiencing the system
we're apparently only experiencing our
individual cognition but then
what does it really mean to be
experiencing individual
cognition anyone else want to have a

thought on that
and also i'll just put up the third
question is
what would be something that you'd like
to clarify or
explore in today's conversation
so shannon
i'm really curious about um this idea
that the mind is skull-bound
while also giving cognition um to
extended systems that like extend beyond
the mind
especially in like being in a
neuroscience
lab and like what i am interested in
studying and explaining is how the brain
is involved with some cognitive process
um while also
understanding and not minimizing the
role that the body or the environment
plays
agreed um within the markov blanket of
this
conversation that we're having there's
some sort of distributed cognition at
play but
who or what is experiencing that and
what are the dynamics of that
type of cognition does anyone else want
to have any
thoughts on these questions or we can
just
start going through some of these quotes
and ideas
okay i'm just gonna go into it
so here we are in multi-scale
integration
beyond internalism and externalism a

2019 paper

last week in 5.1 we went through it
pretty broadly today we're going to
follow up on some key quotations some
key ideas go through the figures
again in this sort of spiraling back at
a higher level

wanting to understand what is on the
table what are the implications of what
is being discussed

the goals of the paper just to rehearse
them were to make

explicit this description of the
boundaries of cognitive systems

uh as being multi-scale and so
when we think that the boundaries of
cognitive systems are nested and
multiple

and the implication of this to the
authors is that

cognition has no fixed or essential
boundaries and

where that sort of took me as far as the
slides go

was what were the implications of taking
multi-scale

integration seriously and how does free
energy

principle and free energy minimization
fit into all of this

so first is there anything that people
just want to

raise as something to address or
think about before we jump into some of
these topics

if not just feel free to write it down
and we'll return to it soon

okay for the first topic yes stephen go

ahead

just just one thing that i think might
be good to mention that came up is this
idea of

um how um we privilege

certain levels and maybe then end up
just stuck at that level

like a psychological level or a
behavioral level

or whatever so um

there's an implication i think in this
um

to to think about the levels

which is kind of maybe some way a bit of
an abstract thing

when across all the levels and how do we
actually

privilege in some sort of intentional
way

the levels that are going to be sort of
looked at as the kind of active
inference engines that are most relevant
so just that's something that came up
for me yeah i would

um actually like to follow up and ask
you

we talk about this privilege level of
analysis and

recall dennis noble's paper what does it
mean to privilege one

level of analysis or is that something
that is just

a priori to be avoided or do we want to
privilege larger systems

smaller systems how do we give each
player in the system it's due respect

and where does that fit in with this

vocabulary of privileging one level of

analysis

you know i was i was wondering whether

that's where that active

um dynamic kind of um structure

that's most that that's a play might

come in i was thinking that there's this

the sense that you've got kind of the

the overall kind of dynamics of the

system but there are key places

where um there there's something i think

shannon mentioned this

last week in terms of the certain levels

which are most relevant

in terms of um how the dynamics are

yielding some systemic

kind of act which is appropriate at the

particular spatial

temporal um sort of speeds

or distances which are relevant for the

study or the experiment or the practice

which is being looked at so i suppose

in some ways you have to start to come

back from the other directions at some

point

um assuming you're moving more and more

towards practice and say okay what's the

what is it that the the kind of

intentionality of the researchers

cool and we'll return to this idea about

are we looking at the dynamics at the

causal structure of the system are those

the same thing

here we're gonna talk for a little bit

about

the big philosophical question which is

internalism versus

externalism so i'll start with just a

definition of the topic from

encyclopedia.com

and here it's about epistemology but

there's analogous

dichotomies in other areas they write

internalism in epistemology is a thesis

about the nature of

epistemic normativity or the sort of

normativity that is involved in the

evaluation of cognition

specifically internalists claim that the

epistemically normative status of a

belief is entirely determined by factors

that are

relevantly internal to the believer's

perspective on things perhaps

skull bound as shannon said by contrast

externalists in epistemology deny this

the externalist says that the epidemic

status of a belief is not entirely

determined by factors that are

internal to the believer's perspective

and

that uh with the spy versus spy

to represent that they're sort of um

playing off of each other they don't

totally disagree

these two viewpoints uh we can see

with this graphic from a 2010 paper

by lynnae and steiner they represented

in externalism and internalism and

i just thought this was an interesting

figure because

it showed to what extent

these two schools of thoughts basically

agree they basically agree that there is

an organism

that is part of a coupling device like

the body let's just say

and through actions there's action on
the environment the environment
provides sensations to the organism and
that's the action perception loop
so that's sort of what we agree on
between these two
sides where is there disagreement well
the two
main things on this figure where we can
see disagreement are first off
what the externalist sees is happening
inside of the organism to a large extent
is strategy whereas the internalist
sees the organism as making
representations about the world
and then another difference that's
reflected on this figure
is that perception the wavy line is
in the internalist's view inside of the
organism
it's skull bound it's happening
downstream of the representation
whereas for the externalist the
perception
is coming from the interaction between
the organisms strategies
and the environmental causality and that
was something i was sort of just
thinking well
doesn't perception still happen inside
even if there's an important role for
perceptions outside
and so just looking at this graphic or
just thinking about where you stand at
this moment
would you say that you're an internalist
and or an externalist
just looking at these graphics does one

of them
seem more or less consonant with your
stance on how ecologically embedded
systems
are in the world uh or
does there seem to be something that's
missing from one of these sides or the
other
if and or is a real choice and we can
pick and
then definitely have you both what what
makes you want to take and in this
divide
um um
i feel like i want to answer that
question at the end of our discussion
perfect i i might as well it's why i
threw in the and
and why the spy versus spy are shaking
hands because
the whole topic and the discussion in
some senses is uh remember we're moving
beyond
internalism and externalism this may or
may not be
a false dichotomy and then the question
would be well
why has this false dichotomy if it is
attracted so much attention or tension
what is going to be the framework that
is going to step beyond
internalism and externalism and
potentially provide a satisfying
resolution to all of the similarities
and the differences
that these two models present with so
again the action perception
loop doesn't seem like there's broad

disagreement on those topics
however there seems to be a few key
issues that are
up for debate such as whether perception
happens
purely internally or whether perception
is related to feedback with the
environment
and then whether what organisms are
doing is best thought of as
representational or as strategic
stephen i think also those diagrams
there is also that
piece in the diagram where they kind of
make out that they're directly
causality and the environment they've
got a direct relationship to
you know like because in some ways you
your actions
you could if we see that as really
you've got access to your proprioceptive
signals and your sensations are kind of
your kind of ongoing stream of data that
that that link to the environment i'd
say is a lot
um less i think both of these two
approaches rely on there being quite a
an idea that my action is touching the
environment as opposed to my action
being
a kind of set of proprioceptive signals
which i sort of
am inferring something from and my
sensations are the environment
you know this idea that i see the world
as opposed to
having to piece together a whole load of
non-linear data over time so i suppose

there is a kind of maybe a challenge to
that part
of the um of the two
ways of looking at externalism and
internalism that's been done in the past
shannon i suppose um
that's true too so you said like there's
a sense in which i touched the world
on stephen and i think
phenomenologically there's a sense that
i touch and i experience the world
but if i'm thinking about the brain the
thing that the neuron
perceives is not the world it's it's
the signals that have been translated
into electrical
and chemical impulses and that's where
like
i really want to know how the brain
works i got to be internalist about it
but if i want to know how i experience
the world
then there's a lot more externalism that
you can allow in
good points and also we'll return later
on in the conversation too what kind of
experiments
would be uh differentiating how will we
reduce our uncertainty about the
strengths and weaknesses of these two
options maxwell um i just want to point
out that our agenda
in this paper was sort of to dispel the
isms
of internal and externalism right um
so i mean the idea is sort of that well
cognitive science for most of its uh you
know from its inception to basically the

heuristically like the late 80s uh was
an internalist
uh program and then externalist
approaches started
appearing in the uh late 80s and 90s
you know that emphasize things like
embodiment you know the fact that
cognition is uh grounded
in the operations of a body um
you know and then you saw the appearance
of theses in philosophy like the
extended mind
thesis or the extended cognition thesis
which is a strong metaphysical claim
that says well actually the
the realizers of cognitive processes
uh spill over into the environment so
that if you're going to draw a boundary
around the cognitive system you have to
include
external components environmental
components as realizers of these
processes
and the kind of hard line externalist
position is basically kind of the
traditional internalist position is the
equal partners principle
which essentially says cognition is
always
necessarily a loop between
uh you know internal structures like the
brain
and action and external structures uh
like you know the ones in the
environment that we're interacting with
and uh our view on that was that um
any essentialist position about the
boundaries of cognition which which is

to say any position that says okay well
we can define a necessary and sufficient
set of conditions
such that we can identify whether this
or that
you know is a bit of cognition and
whether it's internal or external to
some you know the organism's boundary
that any essentialist position
of that sort um is going to be
wrong-headed
that there's not kind of a
one-size-fits-all way to
cut up systems so basically we're making
two kind of
ontological claims about the boundaries
of cognition
uh which is that they're multiple and
that they're nested
um so you know they're they're multiple
in the sense that depending on your
uh what you want to explain then the
relevant boundaries uh might be
around you know a cluster of cells in
the brain or it might be the brain
itself or some region of the brain or
maybe it's the brain body system
or maybe you know so it depends um
and then the the epistemological point
which is precisely that it depends on
our explanatory interests
um so you know it may be that um
if you want to explain for example
daydreaming or some other
uh kind of phenomenon well you're mostly
going to be appealing to brain processes
but if you want to explain sport
performance then

this loop thing is going to be the most important and importantly the the the point of the article is that uh the free energy principle gives us reason to think that we shouldn't take these strong essentialist positions uh that there's a way around them in a way of being more ecumenical and flexible about where we situate these boundaries very nice i agree and hopefully that's where this discussion will take people is beyond the dogma towards the specifics of the situation and i i also was thinking about like dreaming versus action uh are these uh internalist versus externalist stances challenged by action list experience like a dream or is a dream are they going to couch that as some type of experience and you know what enough with that let's just look for a framework that will help us deal with the specifics of the situation so speaking of internal and external and how they're treated under the free energy principle let's talk a little bit about markov blankets so three kinds of markov blankets are shown on this slide there's a quote in the paper that the markov blankets are a result of the system's dynamics

and two twin questions that i wanted to raise and think about were what if we compare two systems that have the same components but different dynamics and the sort of inverse of that question which is two systems that have different components but the same dynamics and so especially if we think about the the markov uh boundary shielding internal states so black box around some level of analysis all we're able to see are the emitted states so the dynamics of the emitted states may be similar even though there's two different underlying realities so two different computers different hardware but then they're able to run the same program at the same speed and then the other side of the question would be okay you can have the same computer with the same underlying component tree but then it might be involved in multiple different kinds of dynamics so how does that fit with the markov blanket what what is the markov blanket really tracking is it tracking the inputs and the outputs of the objective system is it tracking the inputs and the outputs of us measuring the system well so one thing to highlight just

before we

we really get started is that i think

too much emphasis has been placed on the

literature in part i mean in my own work

also

on the markov blanket on its own uh

markov blankets are very common

and explanatory uninteresting just on

their own for the most part

uh so i mean a markov blanket you have a

set of random variables

uh you wanna see whether uh you know

this and that one are uh independent

uh well the markov blanket is the set of

random variables that renders them

independent

uh in the sense that uh

basically they're uh

that conditioned on the existence of

this

other set of variables then uh these two

uh

are statistically independent um in

terms of like their

yeah so uh that's cool

but there are markov blankets everywhere

right so you can think of like the

present as a markov blanket between the

past and the future for example

uh it's a very general thing um

in the context uh that we're talking

here

uh the markov blankets like get their

explanatory purchase when you combine it

with uh the other kind of moving parts

of the free energy principle

formalism which are the non-equilibrium

steady state slash

generative model thing so basically the

markov blanket

tells you which set of states

kind of belong to the system and which

don't

and then the non-equilibrium

steady-state density tells you well

what is the basically like the the set

of correlated values that these states

can have

uh such that the phenotype is maintained

and it's the two together

that really get you the explanatory

power of the free energy principle

sort of like where the system is and

what it is with the where kind of

answered by

the markov blanket thing um and the what

answered by uh the generative model uh

aspect of it um yeah and

so the markov blanket itself uh

in the free energy principle stuff you

get it from a

sparsity constraints on the flow so

basically this means like

i know it sounds technical but it

basically means that all of the states

of the system are flowing

right they're changing dynamically

they're moving around

in in the state space the system is

moving around in its state space

and uh basically when we say sparsity

conditions on the flow

it's that some variables don't affect

the the change of other variables so

that

uh you know the rate of change of one

variable

with respect to uh some other set of variables is a relevant kind of way of measuring the influence of one set of variables on another

and i mean if the general idea is if conditioned on some set of variables uh the rate of change of one um one variable relative to the other is zero

then you can say that there is an independence statistically right so the way that the blankets are constructed is by analyzing uh the dependencies structure of the flow and then saying okay well the dynamics themselves seem to carve out

these kind of independencies so it's a dynamical notion and i'm emphasizing that because the dynamics themselves are premised on the whole generative model

aspect of it so it all kind of moves together and

yeah and you don't get the free energy stuff just out of more call blankets

shannon yeah um

like following up on that i think

there's a way to

rephrase these two questions and that the first question

comparing systems with the same components but different dynamics

um is asking sort of how

the same set of components if you give them a different set of initial

conditions can exhibit completely different behaviors or like come to embody different phenotypes in a certain niche and then the second question what if they have different components but they have the same dynamics is a question of multiple realizability where you could have the same sort of computation or something enacted with different components and these questions together i think highlight the importance of μ if you have this markov blanket over time this sort of like system of interest over time the importance of the way that system is structured and the function between μ we could say the different nodes underneath the markup blanket μ and that's more important than the actual physical stuff it's made out of maybe and back to the first question the role of the environment in affecting the internal states of μ assume cool thank you and well stated and tied back to these other questions about multiple realizations the follow-up question to that was just in which ways the blanket is tracking just the pure observations of the system versus the causal relationships and you gave a great example about how depending on the initial conditions that

means that the uh
the conditional dependencies in the
systems may be very different
and so in those cases the markov
blankets are different and in some
senses this is an implication of saying
that the boundaries of cognition
are context dependent because it
shouldn't surprise us that
the markov blanket and we will get to
the generative model and internal states
soon but it shouldn't surprise us that
the boundaries
change because there's times where the
cognitive process is happening
in different causal ways or different
spatial scales
stephen yeah i think also
this might be where that challenge with
the the mind and the environment because
we talk about
what what mental processes or what
environmental
processes but what's the constituents
that are being put together in terms of
like the
morphology or the actual you know am i
engaging my muscles in my hand which
and that has you know the hand is
constructed so it's not just my feeling
of my phenotype
about how i should act it's not like my
knowing of my phenotype it's like
my actual biological
engineering is going to enable certain
things to happen
you know unlike a chimpanzee or
something like that there's certain

things i can just do
so there's this middle bit which um
you know we're able to adapt and adjust
to our cognition and say what we bring
what
gets brought online between the
environment and the sensor
so that that's the bit that kind of
really
complicates things but obviously seems
to be really relevant by
i think sometimes gets lost when it gets
kind of more theoretical
yep let's look at this very provocative
quote which is in this sense cognitive
science might be understood as the study
of generative models and processes
it is the business of modeling the
correlational or causal structure of
actions
and observations of the organism the
generative model then
is not the vehicle of something like
content or mutual information
instead it is the tool that we use to
study cognitive systems
as an explanatory model and indeed
perhaps more speculatively
the guide or path living systems entail
and
follow to stay alive as control systems
so in both of these quotes there were
some
pretty interesting notions from a
cognitive science perspective
and also it uh hints at this
way that the markov blanket or the
active inference

uh idea it not only is how organisms
are in their environment and so that's
referring to
this path that living systems follow to
stay alive
in other words that that's what the
animals are doing or that's what the
bacteria is doing
but also it's the tool that we use to
study cognitive systems
and i thought it was pretty interesting
that it was first and foremost
uh and unambiguously presented as a tool
that we use to study systems
and then perhaps more speculatively it
was
a guide that living systems follow to
stay alive so
yeah i'm curious just about that
breakdown yeah to give you some context
this
this uh this paper multi-scale
integration
um the schrodinger answering
schrodinger's question
and a tale of two densities were all
originally the same paper
uh that we split up into different lumps
uh with another answering short answer's
question being the more kind of
theoretical mathematical statement and
then these two other papers being kind
of philosophical explorations of the
consequences of all this
um so over the past few months it's
become clearer to me
what's uh going on here
and i have a paper on this now which has

been published in entropy with yuneshi
polito and carl fristan it's called is
the free energy principle a formal
theory of semantics
um and yes is the answer to that
question

um and basically uh i've been playing
around with the idea that there are kind
of two two levels of instrumentalism
uh that are at play here so
instrumentalism broadly speaking
uh in the philosophy of science is the
view that well

the work of science uh happens by
leveraging scientific models and
getting them to do uh some interesting
explanatory work uh
explaining the variance in our data and
stuff like that so

uh an instrumentalist would claim that
scientific models
aren't literal descriptions of the
universe but

a kind of useful tool that we as
scientists

use to explain the main causal features
of the phenomena

around us so there's a first level then
that you can be instrumentalist about
the free energy principle and i think
that in you know

on a good day when i wake up on the
right side of the bed i'm an
instrumentalist about
scientific models uh

you know uh science is about i think
you know comparing models of data with
theoretical and formal models

ultimately uh and these can be read in a very very large way

um so you know on a bad day i'm a realist

or i have a realist bent and i really think that these models are getting uh you know something like that uh the true causal structure of reality but that's

that's i think science scientific hubris maybe or whatever

but anyway so there's this kind of meta level uh

this philosophy of science kind of instrumentalism

uh level uh and i think that's there's a debate there

uh you know you can you can some of my uh you know

close friends and collaborators think that the fep is finally a literal theory of you know how brains uh you know process

information so it's not just a metaphor or like uh you know an

as-if thing but it's really this is how it happens

all this is independent from what the the the theory the free energy principle itself says

and so the more controversial claim but i think increasingly i think this is the only way to really interpret what the math does

like coherently uh what the free energy principle

is in a sense is the statement

about how organisms are able to exploit

use or leverage the statistical structure of their bodies in motion to generate patterns of adaptive behavior so from that point of view you know regardless of what your position is at the meta level or the philosophy of science level at the theoretical level what the fep is telling you is that organisms are exploiting a model effectively that they are a statistical model that they embody uh in order to generate uh patterns of adaptive behavior so that was the that's to clarify increasingly i think that you know double instrumentalism i'm all about this nested stuff right so instrumentalism nested within instrumentalism is probably the right way to uh to approach these questions but uh it definitely for sure at the theory level the free energy principle is an instrumentalist theory of cognition regardless of whether we're realist about the ontology that it might entail thanks for that clarification and kind of reading between the citations there and i also really agree with that it's sort of like well we've uh tuned into the idea that it really is about our model of the world and about us in relationship with the world and now people just say well

how is the world really and then the
free energy principle saying well it's
really about the feedback between the
organism and the environment and then
we we still want something to pin down
to

that's really how it is but that's the
instrumentalism of
science is whether you use optimal
foraging theory or an economic
theory or some other theory that doesn't
change what the
system is doing the bacteria doesn't
know doesn't care

what version of which theoretical
framework you're using or
you know what what papers you you have
or haven't read it's doing what it does
and the free energy principle is saying
that what it does is as you had just
described

uh leveraging the bodies in motion
they'll also have some questions about
where thought

fits into that um and that's really it
it's really us in feedback understanding
our environment to guide skillful action
and uh it you can't just cut the loop
there and say well then what is just
what is really happening it's really
what's happening and the real
implications of that

are as you described so if there's no
other

comments on this slide i'm gonna go to
this next

few points about what is internal and
what is generative so first

was a very interesting quote in the latter part of the paper and they wrote first we take issue with the claim that under the free energy principle the generative model is something internal to the organism so they're dissenting from the opinion that the generative model is something that is merely internal i.e that the generative model comprises neuronal vehicles or any other vehicles i guess like cars rather the generative model is a mathematical construct that explains how the quantities embodied by the system's architecture change to transcribe that is update beliefs about the causes of the system's sensory observation what should be at stake in the debate between the internalists and externalists is the status of the guess that the organism embodies namely the posterior beliefs encoded by internal states and whether this guess does or does not constitute the limits of cognition understood as the avoidance of surprisal or informational homeostasis so my question here was if the internal states are not generative models or another way to say if the generative model is not something that's just internal to the organism what are the internal

states

um and then the other

side of that question is well then what

or where

is this generative model playing out

so i i can i can answer that i think

it's it's a lot simpler than it might

seem initially so

in machine learning if you have a bunch

of variables

uh some of which are hidden some of

which are observable

and you write a joint probability

distribution over all of those variables

then that joint probability distribution

or density

is called age a generative model

so it's called generative because uh

by representing all of the statistical

relations between

all the different variables as this big

joint density

you're effectively able to generate

the the outcomes or the data

that you would expect you know

contingent on

these causal relations actually holding

between

all of the different variables uh that

you're

talking about so just from a

mathematical point of view

this joint density is never represented

explicitly anywhere in the brain

in the free energy active inference

approaches uh

typically this whole joint density in

all of these kind of

more formal uh models
what you'll typically see is like one
big probability distribution written on
the
left hand side of an equation you know p
over
all of the states that you're trying to
look at often these will be states and
precisions and
you know all these things and then is
equal to
some big product of likelihood and
priors
uh and so what's actually encoded in the
brain
uh has to do with these uh likelihood
likelihoods and priors so the the idea
is that
there is a generative model at play
but it's only at play in the dynamics
the generative model is sort of like
uh the the wave of falling dominoes
um the wave itself is only present in
the kind of
you know motion or dynamics of the
domino pile falling
uh yeah so what we're kind of proposing
here and
we'll be discussing this a bit more in
the next week's uh
with a tale of two densities the paper
which is specifically about this
but the idea here is what we're what the
body
what body states actually encode
are posterior estimates right
so uh posterior estimates of the value
of states like your your body is

basically your best ongoing guess
as to what's causing uh your sensory
states
and yeah the generative model itself uh
is sort of like the point of reference
for the dynamics
uh so i mean if you think about it what
is it this joint probability
distribution thing
well it's essentially a kind of surface
that uh the system is moving
around over or the descript equivalently
that describes the probability of
finding the system in a given state at
random if you're just sampling it
and what it's telling you is the
allowable covariations between all of
the values
of the variables that make up the system
such that the phenotype is maintained
so you know uh speaking a bit loosely
perhaps
uh this generative model thing just is
the phenotype of the organism in this
kind of broad statistical sense of
harnessing all of the possible
uh you know combinations of states
that are compatible with the continued
existence of the organism
um so i mean it's a bit it's a bit um
it's a bit uh hard to grasp i'll admit
uh but at least the mathematics of it
kind of look like that
thanks for that explanation and it's why
we're here is to really
dive into some of these key issues i
think just
reflecting on the pieces that um i'm

taking away from this
is one is that the free energy principle
is describing how scientists investigate
systems or investigators
of any kind and then um as was put
perhaps more speculatively
is what is actually uh happening in the
system
but in some sense that's secondary
because it's so important
to be clear about how we're approaching
the system so instead of saying oh well
it's an optimal forager
it's like no i'm studying the ants
foraging within an optimal foraging
framework
no one can take that away from me and
then whether the ant is doing optimal
foraging or some other thing
it's almost a secondary question because
it's so critical to be clear about our
perspective
and then the second piece is this
internal and external
debate or whatever it is tension between
those two
stances is to be clear about where the
generative model
and um is and what it is and isn't
and also it's just a map in territory
distinction so
let's look at a few more quotes from
this paper
here's a quote the generative model is a
statistical construct
aka that means by people that
transcribes the expected sensory causal
regularities in the process

generating sensory states the generative model

is used to model the set of viable phenotypic statistical relations such as preferences and action policies that must be brought forth by the organism in active inference in short a model model of a viable state of being for the organism

through active inference internal states are tuned and this tuning changes its posterior belief

and hence the organism's best guess about what caused its sensation that usually include its own actions in other words a generative model can be used to understand how organisms are able to track or infer

their own behavior and one key part there is that the best guess doesn't mean the one that's closest to reality per se

it means it's the most action oriented or evolutionarily relevant guess so the best guess for a loud noise isn't just doing inference on the location of the noise or on the exact uh type of object that caused it

the best guess when you're doing inference on policy is actually about how the body should change its dynamics and then the um

question that i raised here is just how does a multi-scale generative model work in the brain or in a social group

and can there be
a generative model without action or
behavior
or external states being influenced so
heavily
maybe things these are things that we
can return to later if no one has any
comments now so yeah stephen
one thing i'll be interested in is if
you're looking at someone's generative
model
like what's the best place to look at it
like in a way if i was to look at
someone's
physical states and how that's
compatible with their existence in the
environment
is that a good read on their generative
model
as much as looking in the brain you know
like
if the states that i'm in constitute
my generative model's best guess of how
i should be in the world
in a way that's a that that is a window
into
what my organism level
cognition is i don't know whether
thinking is the right term
oh yeah i i agree with that i think well
if you look at a lot of the
the active inference modeling uh that's
based on the partially observable
markov decision process um you know
implementation
um yeah i mean for the most part it's
the behavioral modeling framework
so what you're trying to do when you're

tuning your generative model is to get your behavioral profile to align with some data that you're generating right so uh you know behavior on some kind of task of some sort uh yeah and it's the the the connection to the the neuro stuff is a bit more elaborate uh like you know it it we rely on like sort of auxiliary hypotheses and other work that kind of suggests that oh these precisions are being estimated here these prediction errors are being generated here and so on so like if you want to you can use a fitted generative model to make predictions about the kind of neural substrates that you would expect to be engaged in this or that task and that can help guide you know fmri and eeg experimental design to kind of probe into like specific brain regions that you think might be involved so you get into this kind of circular experimental design thing if you want to do the brain aspect but actually these techniques are more easily used for behavioral modeling because you don't have to make all these assumptions about brain structure yeah it's just one comment on there and this was clarified by reading the spm statistical parametric mapping textbook

the generative models were used
extensively to
rather than just do descriptive
statistics on the very
complicated data structure of
neuroimaging
what was being done in the spm approach
was a generative model of brain states
was being used
as a generative model for the data and
so again that's that
instrumentalism of the scientific kind
that just from the purely
statistical like how am i going to write
this paper perspective
having a generative model or having the
brain regions connected in a generative
model
that generates data given certain
hypotheses about how brain regions are
connected
that turns out to be a lot more
tractable to study and it allows you to
do model comparisons
in a very very direct way and then
maybe the activity of what those brain
regions are doing
is itself making a generative model a
cartesian theater or something like that
it'd be wild and amazing if true and
also wild and amazing if the
bold the brain oxygen level dependent
signal captured that
but at the very least and again this is
sort of retreating to the
the scientific instrumentalism at the
very least it's a way that we can
investigate

the changes in the dynamics of those systems and so absolutely i mean this is the duality of techniques that we use right on the one hand there's dynamical causal modeling which as you're pointing out dan doesn't assume that the process that you're modeling is itself an active inference process although it could uh so you know this is the technique that we use in fmri data analysis for example and it's also the technique that has been used recently by uh carl friston's group to model the spread of coronavirus uh which you know you could think that uh you know maybe uh corona itself is an active inference type thing but you don't need to make that assumption to use this modeling strategy and try to extract you know using the what the fmr what the corona studies do essentially is take a bunch of raw data like uh you know raw number of uh of infected a raw number of deaths from corona and use that as data that you're essentially trying to uh then write a generative model of right and fit the generative model to the data the time series effectively uh to kind of extract uh the causal structure of the disease process in this case that's

leading to the data that we're uh that
we're observing
yeah active inference modeling can be
combined with this
but makes the assumption that the agents
that we're modeling
are all performing a kind of active
inference uh
variational inference thing nice
stephen i think this also links nicely i
mean i do a lot of
uh with spatial cognition and how so we
think about oh i see something well i
see something within this direction
there's actually a recent paper i saw
where
people who've lost their here who are
deaf basically
um they're the brain region um
for the their vision it actually maps
out wider
and has less resolution in the middle
because they no longer can hear things
coming in from their sides
and they can't perceive things by
listening to something that comes into
their auditory field so they have to
expand out
their visual field and so this this
this sense of like how we bring in our
sensorim
because we can hear all around us but
only for a certain distance we can see
a long way but only in a certain
direction and
it could somehow there's another page
heard about but people will touch their
nose it could be because you're trying

to think through smell
it's like you know like you sniff out a
problem so you've got this kind of like
the modality like just when you try and
feel your way into a problem and ground
yourself
you don't ground yourself with your eyes
right you ground yourself with your
bottom or whatever it is and um you you
hear
the field or you listen to the field or
you
see the future or you sniff out
wrongdoing
or you taste there's a bad taste in my
mouth
about so this this is kind of
interesting because
these these are kind of mechanisms
that are kind of just pure sensorim but
there's also this kind of um
the way though sensorium relates
spatially into the environment
and couple um
you know what sort of statistical
parameters are they
relating to so yeah i thought that kind
of fits in with what you were saying
yep thanks for that and just to give one
last comment on this generative versus a
descriptive model
um what reality hands us what the csv
that it hands the scientist or the
images that it hands us from our sensory
observations
we could go just from the data and just
identify
keep on going upstream until we have a

model of the world
and that's sort of correlation all the
way down you can say well it was big
and red and you can keep on carrying
that correlation between bigness and
redness
but it's always just going to be
correlational and then the other
approach
would be to go this back and forth
between the observations
and then an actual generative model
where let's just say big and red
cause each other or there's a hidden
cause that causes big and red to exist
and then through expectation
maximization em algorithms or other
approaches fitting that generative model
gives us a lot of power and it's very
computationally tractable
to fit um the causal relationships
that could generate the observed data
and so the expectation maximization
algorithm is a good starting point for
people who want to learn more about this
in the scientific instrumentalist
perspective
for the next um yep alejandra
oh yeah um i was i was thinking
um there's
i don't know how to say the maybe it's
correct to say a boundary
between this um autopaetic
process in thinking in a cell for
example
when we can talk about a generative
model and
allostasis processes

on the other sides like
only homostatic processes
so when you can
um like observe a cell and okay this
this
this how to poses um
processes um are
driven by a generative model or
they are just like a more reactive
and homostatic
i i don't know but i was
all the week long i was um
thinking if if there is like a boundary
or any distinction or
just to make a an inference about
metabolic
changes uh in order to maintain these
um like desired states or
just react and compensate them um
i don't know but i'm quite confused um
um when taking this autoprojective uh
theory about uh
living things and and then thinking
about
um generative models inside this
uh theory so
i i don't know if i'm i'm clear in i
thought
i was pretty clear i mean um so i should
like you know provide the the caveat
that
uh most inactivists i think
vehemently disagree with me about this
uh
but uh you know my hot take on this
is that uh the free energy principle
completely
uh kind of subsumes

uh autopoietic and activism meaning that
everything that an autopoietic and activist
can do within their framework we can do
better

uh and the reason for that is that uh

well so autopoiesis

right is self production and my

understanding at least of this

for having i consider myself a kind of

reformed and activist this was like my

main thing for a while until i

i stumbled upon the free energy

principle uh

so an activist kind of start from the

idea

of autopoiesis autonomy and then kind of

work out

uh like how it works and uh

what you can say on the basis of uh

you know autopoiesis uh

so the free energy stuff is cool

i think because it explains to you how

the autopoiesis gets established to

begin with

i mean i haven't written about this but

i kind of feel like

like what active inference is doing is

putting the

adaptive loop first right rather than

just

saying hey look there's this markov

boundary

uh you know like an activist would hey

look there's this autopoietic boundary

that's kind of producing itself

like the active inference framework is

asking so what are what what is the set

of processes that have to be in play

such that a boundary can be maintained
over time
and then the generative model is is
really just kind of
it's a probability density over all of
the states
that are kind of uh secluded behind the
markov blanket
and that that basically tells you okay
well so what are the values
of these states behind the blanket that
are allowable so look at
a very simple uh example of this is
uh so let's consider my the boundary of
my skin
you know as the kind of organismic
boundary for a second
well one of my internal states is core
body temperature
and my generative model has a kind of
normative density
over this state which says 36.5
degrees celsius with a very small
variance
and uh you know when i'm detecting a
discrepancy
from that kind of preferred data point
i initiate action to bring myself back
to that preferred data point so you know
uh it's getting cold now so like you
know i might put on a parka
if i'm uh if i'm trending in the the
kind of negative direction
uh i mean the point about allostasis and
homeostasis is interesting i
i think it's it's it's it's increasingly
been worked out by people like andrew
corcoran and

uh jakob howie uh but i mean like you know homeostasis is all about what i just said right so you have a preferred data vector and you're trying to uh get yourself as close as possible to that data vector uh where that data vector is is often going to be something like okay well my body temperature or whatever uh allostasis is about the control of that data vector especially in response to stress and environments that are basically frustrating your your desire for certain kinds of data and uh i actually have a paper that we've been preparing for like a year and something now with uh irena or arnaldo and uh casper hess and andrew uh precisely on uh on this issue uh the the paper is about a depression as an allostatic uh disorder uh so the idea that you know if you need to continuously uh be involved in an allostatic process where you're you're always away from your kind of default uh data distribution that you would prefer then it leads to a basically a kind of inflammatory condition where you're kind of accumulating a chronic chronic inflammation effectively from having to stay far away from your uh preferred kind of homeostatic set point

yeah um just adding
something about that and thank you
maxwell actually
i was well i am now a reform and
activist
also and thank you for that paper
the two tail uh densities um
thank you for reading it a lot yeah yeah
um
but um i don't know what you all think
but um is there still any space
um to reactive process processes
without um inference
maybe i'll spirit to the metabolism of
uh angle on the reactive and on the
predictive coding
uh side or not sorry not predictive
coding just predictive modeling
so we can think about like and i just
quickly
got the one second response to this is
like well
uh inference here just means tuning
statistical structure
so it's nothing like uh you know
that paper a tale of two densities tries
to make this point i think like
the the free energy formulation doesn't
really it's a bit counterintuitive but
it doesn't necessarily require you to
explicitly compute and represent a
prediction error
in the way the predictive coding might
for example uh
if you look at the math of it the free
energy only ever
exists in the dynamics and by that i
mean the updates

right like so you really just need a
target data structure and
some sensory data and that's enough to
engage in active inference if you're
able to compute
the difference between the two uh yeah
and then then it's just gradient descent
like if you have a mechanism that's in
place
that can track and keep a like a lid on
that discrepancy then you're engaging in
active inference
uh like it doesn't need to be
like this explicit kind of computation
so uh
you know from that point of view the i i
i i'm always kind of shocked at the
response of an activist uh this kind of
framework because i think it's
everything that an activist would want
right like
it's it doesn't involve explicit
computation uh
whatever information theoretic
quantities are involved only exist in
the dynamics i mean
you know
nice um in just that last closing point
on the metabolism and also to tie it to
semiotics so a sugar molecule touching
the tongue
it's through a receptor which transmits
through neural signaling
that sweetness and that might elicit in
a reactive way
the secretion of insulin but also
insulin
gets secreted in advance of a meal as

well as in a circadian
rhythmic pattern that is adaptive over
evolutionary time and so there's this
meaning making
of the sensory inputs of the the
sweetness of sugar
isn't necessarily related to uh
impending spike in fact
artificial sweeteners are example that
the sweet
taste or the activation of that receptor
doesn't necessarily
entail a subsequent increase although in
our evolved history it did
and so there's space in that whole
mechanism with the tongue and with the
blood sugar and the pancreas
there's space for reactive integration
of stimuli
through semiotics as well as for the
predictive
secretion of insulin in advance of a
meal um i'm going to go
to the next slide because we're coming
along pretty nicely
so here i just found some funny you know
memes about changing your mind and mind
over matter
and stuff and the questions i
put on here was first with a quote and
then two questions
the quote was under the free energy
principle
the system's posterior belief is refined
or tuned under the generative model
through a process of
variational approximate bayesian
inference and it becomes a tight bound

on the true posterior belief it aspires
to
and so this sort of aspirational way of
phrasing what
systems are trying to do it's not just
that it wants to uh
change its body temperature to be
healthy it aspires to it um
i was wondering what people thought
about how might these bayesian posterior
beliefs be related to
consciously held beliefs or thoughts and
then
where might variously things like
affirmations
meditation education or propaganda
fit into this so all of these involve
sensory as well as thought processes
how do we figure that these bayesian
beliefs which may or may not be
consciously held how does that relate
why can't we just change our mind to
understanding the free energy principle
or
change our mind to be different
and these memes sort of suggest that it
is possible
and there's entire industries built
around telling people if you
can change your mind you can change your
life but
what actually enables someone to change
their mind so that they can
change their life and how should we
think about that in this
multi-scale cognitive framework
stephen this actually ties quite nicely
into some of the work

around mental space psychology um
because if you've got a metaphor
like a metaphor can be a construct in a
way
which is a bit like the body but we have
like a mental
model of how things fit together and
if that metaphor is still the same i can
update my beliefs fairly quickly right
i can if i'm still in my kind of like
scientific
chemistry paradigm or whatever i can
update my beliefs the challenge is when
i have to
transform or update my paradigm
and re-integrate into a new
metaphor so you've got this
this in the way like you're saying here
this so this this idea of being able to
do approximate bayesian influence
kind of is quite quick and quite easy to
do
if you're recalibrating posteriors and
sensory data and that
into you know like the sense that like
we're all working together you know and
we're all going to try and work this
team together and then if it shifts to
being like
i'm the boss now i've never been a boss
before suddenly everyone has to do what
i need to
tell them now suddenly my whole mental
model of what it means to be at work has
shifted
now that can be something then that i
have to do like
doctors you have to learn your practice

you have to re-learn
what that means because you have to
reconfigure
and then you can start to work at speed
again so i think this
this is quite quite a nice way to tie
that together
yeah these are really just um thoughts
to raise we'll
move on but just i i wondered okay well
if the body
is some you know if my arm is aspiring
to be this length or aspiring to have
you know this ratio of muscle to fat or
something like that
where do our conscious beliefs fit into
that and are those upstream
of our phenotype are they downstream are
they themselves a phenotype
how do mental states influence our
physical
states and action i think it depends
what you mean by conscious
what do you mean by conscious also look
if by conscious you mean
a form of like reflexive access
uh then what you need there is a kind of
temporally or parametrically deep
architecture
so by that i mean by temporally deep i
mean that you don't want a
just a generative model that anticipates
the next kind of data point that you'll
be
sensing but you want the generative
model that's capable of
entertaining like temporally deep
counterfactual

uh like observations contingent on
courses of action so you might think
that
if a system is able to entertain
you know possible counterfactual futures
contingent on courses of action
then it would be more conscious right
uh than something that couldn't do that
so uh
our preferred approach recently is a
parametric depth we've been exploring
this since um
a paper that actually was just accepted
in neural neural computation
called deeply felt affect
and it so in that paper what we do is
effectively induce a hierarchical model
that is able to make inferences about
its own inferences
so by that i mean uh this model
takes as at a second layer of the
generative model
it takes as its data for inference like
its observable states
the posterior state and precision
estimations at the lower level
so from that point of view what these
systems are able to
to do is have a rudimentary form of self
access
right where they're able not just to
make inferences about what what's
causing their data patterns
but inferences about their own
inferences and how much they trust them
and so like from that point of view that
might also get you slowly closer and
closer

uh to something like consciousness
uh in the access consciousness sense of
like you know having
a kind of nested or reflexive access
to states of your own being if you're
talking about phenomenal consciousness
i think all bets are off i have no idea
no idea whatsoever
uh you know that's a tough one
it's a it's a it's a very hard problem
yep
well it's just things to raise let's go
on
to up alejandra
um yeah i was
reading the this image that if you can
change your mind you can change your
life
um maybe
i i can read this sentence the way
around
if you can change your life you can
change your mind in terms
in this active engagement with
um with the world so if you want to
think differently
behave differently um it's not just a
desire you have to
do it so then considering um
that cognition is relational
so all these mark of blankets and markov
blankets of markov blankets
so going up to your consciousness
so um by bodily interactions
i think is the only way you can change
your mind
and be conscious about it so
yeah sure sasha

yeah i uh like that restatement
of that um that fun phrase if you can
change your life then you can change
your mind
because um that just reminds me
i i mean the reason we have all these
sayings is because it is really
hard to enact behavioral change and um
it shouldn't be easy to change your mind
because then you
wouldn't have learning and memory and
you wouldn't have these
patterns that your life depended on um
and so thinking about uh
what it means to change your mind or
update your priors about something
like if i want to have a different
lifestyle or
exercise more or something like that
it's not just as simple as deciding that
then you actually have to follow through
and so how to make that loop easier
to complete going from changing
your actions to then updating your
beliefs it's kind of like
taking it's like starting to take the
first step before you know which
direction you're moving in
shannon i think this also
brings in um so like vygotsky or psja's
work where you have
your cognition scaffolded by an external
teacher role um and like once
you're developed you're not a child
anymore um the scaffolding role
comes through propaganda or education
or affirmations that are um put forth by
whatever

our sociocultural group is or whatever
our social media algorithms are throwing
at us
um and so it's in as much as
changing your behavior or changing your
life can change your mind also changing
the way
that you interact with the
particular the particular regime of
attention like the particular social
group that you engage with
or that you are
nudged to engage with through
interactions on social media or even
just
regular news media um all sort of
like nudge in less uh volitional ways
your like bayesian posterior beliefs
in terms of like conscious higher level
beliefs about
the world
agreed well said uh it's not enough just
to say i'm the kind of person who runs
every morning
because if you're not running then
you're gonna be uh
not having your sensory input aligned
with your
even stated beliefs so it takes actually
the action
to make that connection stephen
yeah i think this is also like i know if
anyone's you know when you walk into a
cathedral it's like if you want someone
to feel
a sense of awe and start to take their
consciousness up to the heavens
you know walk into a 400 foot high

cathedral

and it will do a lot to you you know and

so it's the way that

and it's not necessarily anyone there

saying you should believe in god

or not straight away anyway

yep let's um go through these last few

slides because we'll also try to get

that figure in

so just some brief notes on free energy

minimization

a quote from the paper was under the

free energy principle

cognition is what the recognition

density or living system does

i.e changing to elude surprises and

maintaining

informational homeostasis by minimizing

free energy

and the way one studies cognition so

there we're again returning to this

double instrumentalism and we're going

to return to this

again in our subsequent weeks

discussions on a tale of two densities

and the way one studies cognition i.e

what the system does is by developing

simulating and analyzing the possible

generative models that explain how the

recognition density of interest the

system of interest

changes so as to attain the minimal free

energy

the mechanics of belief is the only

causally relevant aspect of the

variational free energy

the free energy may or may not exist

what is at stake is the causal

consequences of the action-guiding beliefs of organism and groups of organisms which are harnessed and finessed in the generative model what matters is that organisms are organized such that they instantiate such a prior to guide their actions and it's not a accident that organism and organization and organ are all related and just to show some brief images from a very nice active inference tutorial um here we can see that there's a generative process on the top it's a forest i guess and it is moving through true states but those true states are hidden and so that's whether or not it's real that's what's happening and observations the o are being emitted and that's the partially observable markov model which we again will return to later and then what the organism is involved in are uh changing its own state through time of its generative model with all these caveats and richness that we've been discussing and the actions end up influencing the state or not and then that in the next time step provides new observations that update the model and then just to look for a second at where free energy

fits into this

so that like in each of our discussions

we can kind of bridge from

the experiential components to the

practice to the scientific theory and a

little bit of the math

what is being shown here is that if the

organism

wants to be minimizing its surprise

and maxwell you can fill me in or

correct here

if the organism wants to be minimizing

its surprise which is on the y-axis

and the less surprising the better given

a adaptive

generative model of the world what it

can actually do

instead of just on the left side this

function of expectation

that's the blue line the way to minimize

this

is actually to try minimizing the free

energy

and the free energy is a lot more

tractable to compute

and it turns out that it is strictly

bounded in a way where by minimizing the

free energy

you go quite a long way to minimizing

your surprise because they're very

closely related in a mathematical sense

well i mean we have to keep in mind that

what's going on here is that the

organism is trying to

sample the environment such that

the new data that it's receiving is

consistent with its preferences about

data

so it's not just a retrospective kind of
let's try to fit
a model to this data the it's it's about
a kind of forward-looking kind of
sampling
strategy uh this is why we
in this other paper that we'll be
discussing next week we argue that this
whole framework is inactive
fundamentally uh yeah and
basically you can cast any action
whatsoever
in terms of you know trying to bring
about
a preferred kind of data structure
uh i mean and if you want to like you
know wax a little
philosophical you could even relate this
to a phenomenology
i mean you know i i said i have no idea
how it relates
this active inference stuff relates to
phenomenology but my suspicion
this is pure speculation but my
suspicion is that uh
phenomenal properties kind of correlate
with
the observable states so that it's
really like the
the patterns of observation that you're
receiving at any time kind of feel like
something
and then there's like an interpretation
of that feeling which is more like the
access dimension
um so that that might go some way
towards
uh kind of resolving some of these

issues

uh yeah from that point of view like the organism is trying to get to the phenomenology that it prefers uh which i think is kind of amenable to you know some some cool like properly phenomenological kind of accounts of what's going on i mean you know dolores makes this point um

and uh i forget which one of these but you know like what is desire right well desire is the desire for a certain constellation of things right it's not just like for a thing at least in there's this thing like it's it's like you know you're you're looking for like you're looking to inactive inferences you're you're looking to sample a data structure that's sort of consistent with the way of

being that you are you know uh yeah yeah and if you uh people want different things and yes agree it's a constellation not just obtaining that singular object

stephen yeah i could ask you a quick question maxwell i was wondering is there like

a shift from like active inference dynamics to

as a kind of a way of thinking to active influences sense making

like the sense like it's an active influence all the way up and sense making

all the way at the organism level or is

is there a transition when you'd use
it again it depends whether you mean
this more kind of explicit sense making
or this more if by sense making you just
mean like
in like interacting intelligently
and by that i mean like adaptively
appropriately or whatever
with an environment then it's all the
way up and all the way down in effect
like
cells and you know cells in a body
and people at a at a dinner table are
trying to do the same kind of thing
um i think for the more kind of
reflexive
explicit forms of sense-making that
you're concerned with in your practice
uh you would need some kind of layered
system so
uh sense making in that sense has more
to do with our narrative stuff that was
discussed a few weeks ago
uh it has more to do with creating these
kind of overarching
inferential archite structures that make
sense of your phenomenal flow
effectively
yeah actually that makes sense in the
sense that i've one thing that bugs me
at the higher levels or
not higher levels but these is people
just use the word sense making
like they're using media at the moment
to talk about
wiki problems and all of this but what
does sense making mean when it's
not an actual thing that someone's doing

so like you either have to say like
narrative sense making okay i'm taking
narratives i'm making sense of it
i am you know tends to just be this word
now that's become trendy but it
oh yeah and in fact some inactivists are
even pushing back on it to say that well
it doesn't really mean very much
uh dan hutto and eric meehan have these
really nice critiques of like the
sense-making paradigm
sasha um my follow-up question to that
is
what isn't sense making oh
uh well again it depends i guess on how
uh we want to define it um
so basically active inference
describes more specifically the behavior
of systems that are actually driven
in some non-trivial way by uh
like the beliefs that they encode about
an external environment
and you might see this as a continuum
thing but
i mean there's a sense in which like if
a rock is trying to predict its
environment
and act in a way that you know minimizes
this
divergence between well it's not a very
good model right
i mean it doesn't do very much uh
so i mean this might be a graded thing
in the sense that like you know a uh
like things that are not sense making
are
things that don't seem to reflect this
kind of

intelligent you know adaptation
to the constraints of an environment
like a rock
right uh but you know for for
people like stephen who means something
very specific
and i think like more kind of
psychological and meaningful and loaded
uh if that's what we mean by sense
making then the vast majority of
what we would call sense making under
the other other definition doesn't fall
under it
so like uh you know you really need a
very specific
kind of system that's able to do engage
in a very specific
form of inference unfortunately i'm
gonna have to uh
to leave at this point thanks maxwell
see you soon thank you very much
everyone this is very fun
bye yeah steven i thought there
yeah just um the the
also there's this sense of sense
breaking or sense finding
so it could be that sense making is is
okay this is
and it works makes sense at the the
shorter time scales okay
participatory sense making is like this
ongoing knowing how to act
but then when you get to these higher
levels like
sense making really is more like um
you've made there's retrospective sense
making
which is really what ends up being what

people think about sense making at scale
is i how do i make sense of how i should
have acted
but if you can get some interest that
normally is it
because so that either means someone's
just doing something habitual
or tacit knowledge where they're in the
flow and you're trying to unpack
the sense making that allowed them to do
that but that's not accessible
because you're so in the flow so you
need to have some form of sense breaking
where some and then there's a sense
finding which i would say is more like
active inference because you're
re you're more consciously trying to
repurpose your
posterior beliefs and your sensory data
to try and assemble
some sort of um understanding of
what's even going on um
it might be that point of stuckness you
know they talk about stuckness so
stuckness is um once you've
kind you're trying to get to sense
making but while you're stuck
um you may need to go off and do a bit
of foraging um
so it's interesting cool let's just
go through these uh last few thoughts
and then we'll just go to
figure six because we had talked last
time about seeing it
so on this slide we are going to be
asking
are there limits on the kind of
cognition that we can study as humans

because
of our own individual cognitive
processes and
where does collective cognition play
into this
which systems are we in a position to
understand and how would we know
how would we do meta cognition as
individuals or groups
about what approaches uh for which
systems will be valuable
and then in the future experiments and
in the implications
of okay we've moved beyond internalism
and externalism all these other isms
what is that going to mean for our
practice as scientists or as
investigators of
cognitive systems and what would be the
implications
of knowing i'm just going to raise them
just so we can get them on there but
maybe for another time or in the closing
thoughts
and then there was this idea in the
paper
the relevant boundaries of cognition
depend on the level being characterized
and the explanatory interests that guide
investigation so we can return to this
general question where are the relevant
boundaries for
cognition and how would we know if we're
too internal or too external
and then in the paper they wrote in
other words
drawing the bounds of cognition means
defining the recognition density of the

system of interest and identifying a
generative model that explains changes
in that system
that follow variational bayesian
inference

and i believe it was sasha who asked
isn't this the same argument
that is used for holism or reductionism
for example uh they'll say something
like

we'll know when we've gotten to too
large of a scale of analysis or too
small of a scale of analysis because our
model will break or it'll be inadequate
or intractable

so how do we get beyond that paradigm of
we'll know it when we see it or

can we ever get beyond the paradigm of
we'll know it when we see it

guided by a certain framework because we
are in feedback with our models

so just things to think about

and then i think let's just briefly look
at figure 6 and then have some closing
thoughts so

figure one was about mark called
blankets figure two

went into the action perception loop and
introduced some of the equations
of fep those were unpacked a bit in
figure two b

figure 3 was operational closure in
graphs

figure 4 was a schematic illustration of
auto

catalysis which is like autopoiesis for
metabolism

figure 5 was that fractal markov blanket

showing that as you zoom into the
broccoli you see that it's markov all
the way down
and here we are in figure six
so if anyone wants to have any comments
on six because we had
uh said that we were going to uh
approach it in this discussion
they can just raise their hand but the
way i would walk
through this figure first is by reading
the caption
and what the caption suggested was that
it's going to be
something like a developmental
morphogenesis scenario
so these are not naive cells it's really
important to take that evolutionary
perspective like
the organism is a arrangement that's a
developmental outcome
that isn't just an arbitrary
developmental state it's one that has
lasted through many generations been
fine-tuned acutely
so these aren't just naive little cells
wandering around
spontaneously um coming to this
arrangement that's shown
in figure f um
it's uh rather a very carefully selected
set of expectations about which
extracellular target signals
like diffusible morphogens the cell
expects to see and so it knows that if
it just pursues its expectation
of the target signal extracellularly it
will be

taking part of a larger developmental outcome

that is evolutionarily adapted so the cell doesn't need to have a cognitive awareness of that

but we can think about this like an anterior to posterior gradient of some diffusible morphogen and what's happening each of these traces on figure a and b and c are through time on the x-axis showing like what one cell ends up experiencing whether it's its expectation of a uh morphogen again that's the sensory input in this model

or its action reflected by its ability to move

then in figure d we see the free energy of the system first slightly rise

but then immediately start converging down down down down down which reflects overall that there's an increased precision

in the ensemble in the aggregate of these cells as they sort of jostle into place and come into an arrangement

that minimizes their surprise about what extracellular target signal they should be perceiving

and what we see is that even though there's no higher level coordination per se

the agents end up converging on this solution that reflects the target morphology in this case it looks like i don't know kind of a upside down light

bulb or something not sure what shape
that would be
but you can see how each of these
different colored cells
is able to be uniquely determining where
its position is like the green cell it
can be thinking something like
i want to be close to blue but far from
yellow and
totally far from red the blue cell might
be thinking something like
i want to have an intermediate level of
the green and the yellow signal
and then the red cells might be thinking
something like
i want to have mostly red signal and
then in decreasing order
yellow blue green and then depending on
what the instantaneous perception is of
those diffusible morphogens
the cells can either update their
generative model of the world
through movement that's the
morphogenesis component
or by changing their generative model
which isn't actually happening like
there's no learning and adaptation
within this model
but over evolutionary time it will end
up being the case
that developmental trajectories that
robustly enable
high fitness morphologies to arise will
be selected for
so that's sort of my short take on
figure six any
comments or thoughts on that
cool just because we're basically out of

time um

i'll just say thanks everyone for

participating and we'll have a closing

thought as well

we're going to provide a follow-up form

for our live participants i'll put that

in the chat in a second

we request feedback suggestion and

questions from

all of our viewers and our participants

and

we just hope that people stay in

communication with us and keep watching

next week and the following week we're

going to be talking about a tale of two

densities

which is the paper that we mentioned a

few times today

and yeah it's going to be a great

discussion next two weeks as well

so if anyone has any closing thoughts on

the discussion

while i post the link in here they're

free to go ahead

if any thoughts otherwise that was a

great discussion cool yeah

thank you for the discussion that was

great awesome

in that case i am going to

terminate the live stream thanks

everyone for participating and for

watching

we'll see you in the next two weeks for

number six a tale of two